

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 4 — Cell Cycle

ANSWER KEY (Version 1)

Total Marks: 65

Time: 2h 30m

Version: 1

SECTION A — MCQ Answers (12 × 1 = 12 Marks)

#	Question	Answer	Explanation
1	The cell cycle consists of two main phases. These are:	C 1	The cell cycle has two main phases: interphase (growth and DNA replication) and the mitotic (M) phase (actual division). G1/G2 are sub-phases of interphase; karyokinesis/cytokinesis are sub-divisions of M phase.
2	DNA replication takes place during which phase of interphase?	B 1	The S phase (Synthesis phase) is when the cell replicates its entire genetic material. Each chromosome produces two sister chromatids joined at the centromere. G1 is growth; G2 prepares for division; G0 is a quiescent state.
3	During cell division, spindle fibres attach to a chromosome at the:	C 1	Spindle fibres attach at the centromere of each chromosome and pull them to opposite poles. The telomere is the chromosome end-cap and has no role in spindle attachment.
4	The longest phase of mitosis is:	D 1	Prophase is the longest phase. Chromatin condenses into chromosomes, the nucleolus disappears, the nuclear membrane breaks down, and the spindle apparatus begins to form — all time-consuming events occurring in sequence.
5	During metaphase of mitosis, chromosomes are aligned at the:	B 1	At metaphase , chromosomes line up along the equator of the spindles (metaphase plate). Two spindle fibres, one from each pole, attach at the centromere of each chromosome.
6	The spindle apparatus of plant cells differs from that of animal cells in not having:	C 1	Plant cells lack centrioles . Their spindle is still formed from microtubules but without centrioles as organising centres, whereas animal cells use centrioles that duplicate and migrate to opposite poles.
7	A bivalent in meiosis consists of:	B 1	A bivalent is a synapsed pair of homologous chromosomes. Each homologue has two sister chromatids, so a bivalent has four chromatids and two centromeres (one per homologue).
8	Crossing over occurs between non-sister chromatids during:	C 1	Crossing over takes place during prophase I . Homologous chromosomes synapse and non-sister chromatids exchange segments at chiasmata, producing genetic recombination.
9	Meiosis produces daughter cells that are:	D 1	Meiosis completes two successive divisions to give four haploid (n) daughter cells . These act as gametes. Mitosis, by contrast, produces two diploid daughter cells.
10	What causes the chromosome number to reduce to half when a cell divides by meiosis?	B 1	In anaphase I , spindle fibres pull homologous chromosomes to opposite poles. Each pole receives only one member of each homologous pair, halving the chromosome number. Sister chromatids are not separated until meiosis II.
11	The chromosomal number of fruit fly is 8. The gametes of fruit fly contain:	B 1	Gametes are haploid ($n = \text{half of } 2n$). Fruit fly $2n = 8$, so gametes carry 4 chromosomes . Meiosis halves the diploid number to produce haploid gametes.
12	The ability of a sea star to regenerate a lost arm depends on which type of cell division?	C 1	Mitosis is responsible for regeneration. It produces genetically identical daughter cells to replace lost tissue. The textbook explicitly cites sea star arm regeneration as an example of mitosis significance.

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) Cell cycle definition and two main phases 3

The **cell cycle** is the sequence of events involving growth of a newly formed cell, replication of its genome, and division into two daughter cells. 1

Interphase: the period between two consecutive divisions. The cell grows, synthesises proteins, and replicates its DNA. Divided into G1, S, and G2 sub-phases. 1

Mitotic phase (M phase): the cell divides into two daughter cells through karyokinesis (nuclear division) followed by cytokinesis (cytoplasm division). 1

OR

G0 phase — permanent vs temporary 3

During G1, a cell may exit the cycle and enter **G0 phase**, where it stops dividing and performs only normal metabolic roles. 1

Permanent G0: neurons — once mature they never divide again. 1

Temporary G0: liver and kidney cells — they can re-enter the cell cycle when needed. Epithelial cells never enter G0 and divide continuously. 1

2(ii) Three sub-phases of interphase 3

G1 phase: cell grows in size, increases organelles, forms proteins, enzymes and nucleotides needed for DNA replication in S phase. 1

S phase (Synthesis): cell replicates its entire genetic material; each chromosome produces two sister chromatids joined at the centromere. 1

G2 phase: cell synthesises spindle proteins, accumulates energy, and forms more mitochondria to prepare for division. 1

OR

Spindle apparatus — animal vs plant cell formation 3

The **spindle apparatus** is made of microtubules (cytoskeleton). It separates chromatids/chromosomes and moves them to opposite poles, equally distributing genetic material into daughter cells. 1

Animal cells: a pair of centrioles near the nucleus duplicates; both pairs migrate to opposite poles and give rise to spindle fibres. Once the nuclear envelope disintegrates, fibres attach to centromeres. 1

Plant cells: lack centrioles entirely. Spindle apparatus still forms from microtubules but without centrioles as organising centres. 1

2(iii) Prophase of mitosis 3

Prophase is the longest phase. Chromatin fibres coil and condense into chromosomes; each chromosome consists of two chromatids (due to S-phase replication) joined at the centromere. 1

The **nucleolus disappears** as its DNA is packed into chromosomes. 1

The **nuclear membrane breaks down** into vesicles dispersed in the cytoplasm. The spindle apparatus is formed. 1

OR

Metaphase and anaphase of mitosis 3

Metaphase: chromosomes align along the equator (metaphase plate). Two spindle fibres — one from each pole — attach at the centromere of each chromosome. 1

Anaphase: spindle fibres pull centromeres, which split. Sister chromatids separate and are pulled to opposite poles as individual chromatids. 1

Spindle fibres exert a pulling force on the centromere, ensuring each pole receives an identical set of chromosomes. 1

2(iv) Telophase of mitosis 3

Chromatids reach their poles and uncoil/lengthen to reform **chromatin fibres**. 1

The **spindle apparatus disintegrates**. A new **nuclear membrane** reforms around the chromatin at each pole. 1

Nucleoli reappear in both new nuclei, completing formation of two daughter nuclei each with the same chromosome number as the parent. 1

OR

Cytokinesis — animal cell vs plant cell 3

Cytokinesis divides the cytoplasm into two daughter cells; it begins while telophase is in progress. 1

Animal cell: cell membrane invaginates at the equatorial region forming a **cleavage furrow** that grows inward until membranes join and separate the two daughter cells. 1

Plant cell: spindle fibres form a **phragmoplast**. Golgi vesicles accumulate at its centre, forming the **middle lamella** and primary cell walls of daughter cells. No cleavage furrow forms. 1

2(v) Four significances of mitosis 3

- **Genetic stability:** daughter cells have the same chromosome number and genetic information as the parent. 0.75
- **Development and growth:** multicellular organisms grow from a single zygote by continuous mitotic divisions. 0.75
- **Cell replacement and wound healing:** worn-out cells (e.g., skin, digestive tract lining) are replaced by identical cells via mitosis. 0.75
- **Regeneration:** sea star regenerates a lost arm by producing new cells through mitosis. 0.75

OR

Regeneration — three organisms and body parts 3

Regeneration is the ability of some animals to regrow body parts removed accidentally. New cells are produced through mitosis. 1

- **Sea star** — regenerates a lost arm. 0.67
- **Earthworm** — regenerates its head. 0.67
- **Salamander** — regenerates its limb. 0.66

2(vi) Cancer, benign tumour, malignant tumour, metastasis 3

Cancer is a disease of the cell cycle in which cells lack a proper control system and divide excessively, forming an abnormal mass called a **tumour**. 1

A **benign tumour** is an abnormal mass of normal cells that stays at its original site and does not spread. 1

A **malignant tumour** is cancerous and spreads into neighbouring tissues and to distant body parts. This spread is called **metastasis**. 1

OR

Meiosis definition and contrast with mitosis 3

Meiosis gives rise to four daughter cells each having half the number of chromosomes of the parent cell. It takes place only in cells involved in sexual reproduction. 1

Divisions: meiosis involves two successive cell divisions (meiosis I and II); mitosis involves only one. 1

Daughter cells: meiosis produces four haploid (n) cells; mitosis produces two diploid ($2n$) cells that are genetically identical to the parent. 1

2(vii) Synapsis, bivalent, crossing over, chiasma 3

Synapsis is the pairing of homologous chromosomes during prophase I, with centromeres in the same position. 1

Each synapsed pair is called a **bivalent**. Non-sister chromatids join at points called **chiasmata** (singular: chiasma). 1

Crossing over is the exchange of segments of genetic material between non-sister chromatids at a chiasma, resulting in genetic recombination. 1

OR

Events of prophase I and why it is the longest phase 3

During **prophase I**, chromosomes coil and condense. Homologous chromosomes undergo **synapsis** forming bivalents; crossing over occurs at chiasmata between non-sister chromatids. 1

By end of prophase I, chromosomes are fully condensed; paired chromosomes repel but remain attached at chiasmata.

Nucleoli and nuclear envelope disappear; spindle apparatus forms. 1

Prophase I is the **longest phase of meiosis** because synapsis, chiasmata formation, crossing over, and full chromatin condensation are all complex and time-consuming processes. 1

2(viii) Metaphase I and anaphase I; difference from mitotic anaphase 3

Metaphase I: bivalents align at the equatorial plate. One member of each homologous pair receives a spindle fibre from one pole; the other member receives one from the opposite pole. 1

Anaphase I: spindle fibres pull homologous chromosomes to opposite poles. Each pole receives a haploid set of chromosomes (still as two-chromatid units). 1

Key difference: in mitotic anaphase, centromeres split and *sister chromatids* separate. In anaphase I, centromeres do NOT split; whole *homologous chromosomes* (each with two chromatids) are pulled apart. 1

OR

Telophase I and cytokinesis I 3

Telophase I: chromosomes reach poles and uncoil into chromatin. A **nuclear envelope** reforms at each pole around the haploid chromosome set. 1

Cytokinesis I divides the cell (by cleavage in animals, cell wall in plants). 1

Result: **two haploid daughter cells**. Each chromosome still consists of two chromatids that are not genetically identical due to crossing over — they must be separated in meiosis II. These cells enter interphase II. 1

2(ix) No DNA replication in interphase II; meiosis II overview 3

Interphase II varies in length but has **no further DNA replication** — chromosomes are already duplicated from S phase. Replication would incorrectly double the genetic material again. 1

Meiosis II resembles mitosis. It has prophase II, metaphase II, anaphase II, and telophase II, during which **sister chromatids** are separated. 1

After cytokinesis II: **four haploid cells** are produced, all with new genetic combinations because of crossing over in meiosis I. 1

OR

Anaphase II and telophase II; final result 3

Anaphase II: centromeres divide; spindle fibres pull sister chromatids (now called chromosomes) to opposite poles. 1

Telophase II: chromatids reach poles and uncoil. Spindle disappears. Nuclear envelopes reform around chromatin at each pole. 1

After cytokinesis II: **four haploid (n) daughter cells** from one original diploid ($2n$) parent. All carry new gene combinations due to prophase I crossing over. 1

2(x) Meiosis and maintenance of chromosome number — human example 3

Diploid germ cells undergo meiosis to produce haploid gametes. Fusion of two haploid gametes at fertilisation restores the diploid chromosome number specific to the species. 1

In humans, somatic cells have **46 chromosomes ($2n = 46$)**. Meiosis produces gametes with **23 chromosomes ($n = 23$)**. 1

At fertilisation, sperm (23) + egg (23) = zygote with **46 chromosomes**, restoring the original number. 1

OR

Genetic variations — two meiotic events 3

Variations are differences among members of the same species; they are necessary for survival in a changing environment. 1

Crossing over (prophase I): non-sister chromatids exchange segments at chiasmata, producing chromatids with new allele combinations not present in either parent. 1

Independent assortment (metaphase I): homologous pairs orient randomly at the metaphase plate, so each gamete receives a random combination of maternal and paternal chromosomes. Both events together are the basis of genetic variation. 1

2(xi) Three differences between mitosis and meiosis 3

Feature	Mitosis	Meiosis
Occurs in	Somatic cells 0.5	Germ line cells 0.5
Number of divisions	One division 0.5	Two successive divisions 0.5
Daughter cells	2 diploid (2n) 0.5	4 haploid (n) 0.5
Crossing over	Does not occur	Occurs in prophase I
Genetic outcome	Genetically identical	Genetically different

OR

Cloning and tissue culture — role of mitosis 3

Cloning produces many identical copies of a living organism in an artificial environment. **Tissue culture** uses a small mass of tissue cells placed in agar growth medium with nutrients and auxins to develop into plantlets. 1

Mitosis makes these possible because every mitotic division produces cells with **exactly the same genetic information** as the parent — so all cells in the clone are genetically identical. 1

Example: **potato** tissue is scraped, placed in growth medium, develops into plantlets, then transferred to compost. Other examples: onion, garlic, ginger, grasses (vegetative propagation). 1

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

3(i) Cell cycle in detail — G1, S, G2, G0 sub-phases and M phase 5

The **cell cycle** is the sequence of events from the beginning of one division to the beginning of the next. It has two main phases. 0.5

Interphase (between consecutive divisions): 0.5

- **G1 phase:** cell grows in size, increases organelle number, forms proteins and enzymes, and prepares nucleotides for DNA replication. It prepares for the S phase. 1
- **S phase (Synthesis):** cell replicates its entire genetic material; each chromosome produces two sister chromatids joined at the centromere. 1
- **G2 phase:** cell synthesises spindle proteins, accumulates energy, and forms more mitochondria to prepare for division. 0.75
- **G0 phase:** cell may exit the cycle during G1 and enter G0, where it performs metabolic functions but does not divide. Neurons stay in G0 permanently; liver/kidney cells enter temporarily. 0.75

Mitotic phase (M phase): the cell divides into two daughter cells through karyokinesis (nuclear division) and cytokinesis (cytoplasm division). 0.5

OR

Spindle apparatus — formation, role, and disappearance 5

The **spindle apparatus** is made of microtubules formed by the cytoskeleton during cell division. It separates chromatids or chromosomes and moves them to opposite poles, ensuring equal distribution of hereditary material into daughter cells. 1

Formation in animal cells: a pair of centrioles near the nucleus duplicates; both pairs migrate to opposite poles and give rise to spindle fibres. The spindle forms around the nucleus. Once the nuclear envelope disintegrates at end of prophase, spindle fibres penetrate the nuclear region and attach to the centromere of each chromosome. 2

Formation in plant cells: plant cells have no centrioles. The spindle apparatus is still organised from microtubules but without centriole organising centres. 1

Disappearance: after chromatids are pulled to their poles at end of anaphase, the spindle apparatus disintegrates during telophase, and chromatin reforms. 1

3(ii) Karyokinesis in mitosis — all four stages with centromere role 5

Karyokinesis is nuclear division, forming two nuclei. It is divided into four stages: 0.5

Prophase --> Metaphase --> Anaphase --> Telophase

Prophase (longest): chromatin condenses into chromosomes, each with two chromatids joined at the centromere. Nucleolus disappears. Nuclear membrane breaks down into vesicles. Spindle apparatus forms. 1.5

Metaphase: chromosomes align at the spindle equator (metaphase plate). Two spindle fibres — one from each pole —

attach at the **centromere** of each chromosome. 1

Anaphase: spindle fibres pull on centromeres, which split. Sister chromatids separate and are pulled to opposite poles; each pole receives an identical chromosome set. 1

Telophase: chromatids reach poles, uncoil into chromatin. Spindle disintegrates. Nuclear membrane reforms at each pole. Nucleoli reappear in both new nuclei. 1

OR

Cytokinesis (animal and plant) + significance of mitosis 5

Cytokinesis divides the cytoplasm into two daughter cells, beginning during telophase. 0.5

Animal cell: membrane invaginates at the equator forming a **cleavage furrow** that deepens until the membranes join and pinch apart two daughter cells. 1

Plant cell: spindle fibres form a **phragmoplast**. Golgi vesicles accumulate at the centre and grow outward forming the **middle lamella** and primary cell walls; later secondary walls may form. 1.5

Significance of mitosis (four headings):

- **Genetic stability** — identical genetic information in daughter cells. 0.5
- **Development and growth** — zygote grows into adult by repeated mitosis. 0.5
- **Cell replacement and wound healing** — epithelial cells of skin/digestive tract replaced by mitosis. 0.5
- **Regeneration** — sea star, earthworm, salamander regenerate body parts. 0.5

3(iii) Meiosis I in detail — interphase I through cytokinesis I 5

Interphase I: genome replicates so each chromosome has two chromatids. 0.5

Meiosis I is the reduction division that halves chromosome number: 0.5

Prophase I --> Metaphase I --> Anaphase I --> Telophase I --> Cytokinesis I

Prophase I (longest phase of meiosis): chromosomes condense. Homologous chromosomes pair in **synapsis**, forming **bivalents**. Non-sister chromatids join at **chiasmata** and exchange segments in **crossing over**, producing genetic recombination. By end of prophase I, chromosomes fully condensed; nucleoli and nuclear envelope disappear; paired chromosomes repel but stay attached at chiasmata; spindle apparatus forms. 2

Metaphase I: bivalents align at the equatorial plate. One member of each homologous pair receives a spindle fibre from one pole; the other member from the opposite pole. 1

Anaphase I: spindle fibres pull homologous chromosomes (not sister chromatids) to opposite poles. Each pole receives a haploid set. 0.5

Telophase I and Cytokinesis I: nuclear envelopes reform. Cytokinesis I produces **two haploid daughter cells**; each chromosome still has two chromatids that are not genetically identical due to crossing over — they enter interphase II. 0.5

OR

Meiosis II in detail — interphase II through cytokinesis II 5

Interphase II: varies in length. **No further DNA replication** — chromosomes are already duplicated. 0.5

Meiosis II resembles mitosis and separates sister chromatids: 0.5

Prophase II --> Metaphase II --> Anaphase II --> Telophase II --> Cytokinesis II

Prophase II: chromatin re-condenses into thick fibres. Spindle apparatus is organised. Nucleoli disappear; nuclear envelope disintegrates. 1

Metaphase II: chromosomes line up separately around the equator. Each chromosome is attached by a fibre from both poles (as in mitosis). 1

Anaphase II: centromeres divide; spindle fibres pull sister chromatids (now individual chromosomes) to opposite poles. 1

Telophase II: chromatids reach poles, uncoil. Spindle disappears. Nuclear envelopes reform. 0.5

Cytokinesis II gives **four haploid (n) daughter cells** total. All carry new genetic combinations because of crossing over in prophase I. 0.5

3(iv) Mitosis vs meiosis — six or more differences 5

Feature	Mitosis	Meiosis
Cell type	Somatic (body) cells 0.5	Germ line cells 0.5
Number of divisions	One division 0.5	Two successive divisions 0.5
Daughter cells produced	2 daughter cells	4 daughter cells
Chromosome number	Diploid (2n) 0.5	Haploid (n) 0.5
Function	Daughter cells become somatic body cells	Daughter cells form gametes
Homologous pairing	Chromosomes remain independent 0.5	Homologous chromosomes pair (synapsis) 0.5
Chiasmata	No chiasmata formed	Chiasmata formed in prophase I
Crossing over	No crossing over 0.5	Crossing over in prophase I 0.5
Genetic outcome	Daughter cells genetically identical	Daughter cells genetically different
Further division	Daughter cells can divide again	Gametes cannot divide any more

OR

Significance of meiosis — (a) chromosome maintenance, (b) genetic variations 5

(a) Maintenance of chromosome number in sexual reproduction 2.5

Diploid germ cells undergo meiosis to produce haploid gametes. Fusion of two haploid gametes at fertilisation maintains the species-specific chromosome number. In humans, somatic cells have **46 chromosomes ($2n = 46$)**. Meiosis produces gametes with **23 chromosomes ($n = 23$)**. Fertilisation: sperm (23) + egg (23) = zygote (46) — the original diploid number is restored. In plants, meiosis occurs during spore formation.

(b) Genetic variations 2.5

Variations are differences among members of the same species; they are necessary for survival in a changing environment. Two meiotic events provide the basis for variation:

- **Crossing over:** prophase I non-sister chromatids exchange segments at chiasmata, producing new allele combinations in gametes and then in the zygote.
- **Independent assortment:** homologous pairs orient randomly at metaphase I, giving each gamete a random combination of maternal and paternal chromosomes. Both events together explain why offspring are genetically different from parents and from each other.